# Supplemental Digital Content 1 — TRIPOD 2015 Checklist

**Manuscript:** Physics-Informed Middle Ear Barotrauma Risk for Hypobaric Chamber Training: A Computational Prediction Model Calibrated to the Colombian Aerospace Force Cohort and Externally Validated Against Italian Air Force Cohorts

**Reporting guideline:** Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD 2015) — Collins GS, Reitsma JB, Altman DG, Moons KGM. *Ann Intern Med.* 2015;162(1):55–63.

**TRIPOD type:** Type 4 — development and external validation in independent cohorts.

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| 1 | Title | Identify the study as developing and/or validating a multivariable prediction model, the target population, and the outcome to be predicted. | Title (full manuscript title); Article type line | Yes |

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| 2 | Abstract | Provide a summary of objectives, study design, setting, participants, sample size, predictors, outcome, statistical analysis, results, and conclusions. | Structured abstract (8 O&N headings) | Yes |
| 3a | Introduction — Background and objectives | Explain the medical context (including whether diagnostic or prognostic) and rationale for developing or validating the multivariable prediction model, including references to existing models. | §1 Introduction, paragraphs 1–3 (Kanick-Doyle 2005 review and three shortcomings) | Yes |

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| 3b | Introduction — Background and objectives | Specify the objectives, including whether the study describes the development or validation of the model, or both. | §1 paragraph 4 (development + external validation in two air-force cohorts); Abstract Objective | Yes |
| 4a | Methods — Source of data | Describe the study design or source of data, separately for the development and validation data sets, if applicable. | §2.1 Study design; §2.8 Calibration and external validation (separate paragraphs for development and validation cohorts) | Yes |
| 4b | Methods — Source of data | Specify the key study dates, including start of accrual, end of accrual, and, if applicable, end of follow-up. | §2.8: development cohort 1 January 2010 – 31 March 2026 | Yes |

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| 5a | Methods — Participants | Specify key elements of the study setting (e.g., primary care, secondary care, general population) including number and location of centres. | §2.8: DIMAE hypobaric chamber facility, Bogotá (single centre, 2,640 m elevation); validation in three published Italian Air Force cohorts | Yes |
| 5b | Methods — Participants | Describe eligibility criteria for participants. | §2.8 Source of data and participants (eligibility was operational; no exclusion by age, sex, or prior ear history) | Yes |
| 5c | Methods — Participants | Give details of treatments received, if relevant. | §2.1 (retrospective observational; no intervention); §2.8 §2.5 (no medications administered for the study) | Yes |

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| 6a | Methods — Outcome | Clearly define the outcome that is predicted by the prediction model, including how and when assessed. | §2.8 Outcome (Teed grade ≥ I clinical MEB ascertained by on-duty chamber flight surgeon using otoscopic, tympano-metric, symp-tomatic criteria, within 24 h) | Yes |
| 6b | Methods — Outcome | Report any actions to blind assessment of the outcome to be predicted. | §2.8 — chamber flight surgeons assessed outcomes prospec-tively for clinical care; no blinding to predictors but the prediction-model fit uses aggregate cohort statistics, not per-subject predictions | Yes |

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| 7a | Methods — Predictors | Clearly define all predictors used in developing the multi-variable prediction model, including how and when they were measured. | §2.8 Predictors (descent rate, mastoid volume, RA, PO’, URI temporal state, PET state, chronic rhinitis, de-congestant use, prior MEB history, swallow and Valsalva frequency) | Yes |
| 7b | Methods — Predictors | Report any actions to blind assessment of predictors for the outcome and other predictors. | §2.8 — predictors are mecha-nistic model parameters drawn from published priors and the operational record; no blinding required | Yes |

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| 8 | Methods — Sample size | Explain how the study size was arrived at. | §2.8 development cohort: full operational registry 2010–2026 (n = 7,271 chamber exposures, n = 173 events); validation cohorts: as published | Yes |
| 9 | Methods — Missing data | Describe how missing data were handled (e.g., complete-case analysis, single imputation, multiple imputation) with details of any imputation method. | §2.8 Missing data — no imputation because anchor is an aggregate population statistic; individual-level missingness is absorbed in the cohort prior and synthetic-cohort sampling | Yes |

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| 10a | Methods — Statistical analysis | Describe how predictors were handled in the analyses. | §2.5–2.7 (state-machine encoding for URI and PET; lumped-parameter muscle mechanics; three-threshold hazard) | Yes |
| 10b | Methods — Statistical analysis | Specify type of model, all model-building procedures (including any predictor selection), and method for internal validation. | §2.3–2.7 mechanistic physics model (no machine learning); §2.8 Calibration methods — log-space bisection plus ABC-SMC against three summary statistics | Yes |

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| 10c | Methods — Statistical analysis (validation only) | For validation, describe how the predictions were calculated. | §2.8 External validation — Italian Air Force cohort priors applied to the calibrated physics model without refitting; cohort-mean per-exposure prevalence computed from synthetic-cohort sampling | Yes |

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| 10d | Methods — Statistical analysis | Specify all measures used to assess model performance and, if relevant, to compare multiple models. | §2.8 Model performance measures — absolute agreement of simulated vs observed per-exposure prevalence against Wilson 95% CI; agreement of projected career-3 prevalence; preservation of Kanick-Doyle Figure 3 healthy-baseline trace under a pinned regression fixture | Yes |

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| 10e | Methods — Statistical analysis (validation only) | Describe any model updating arising from the validation, if done. | §2.8 — no model updating during external validation (cohort priors tightened for Italian AF tympanometry pre-screening; hazard constants held fixed) | Yes |
| 11 | Methods — Risk groups | Provide details on how risk groups were created, if done. | §3.3 — per-outcome probabilities (barotitis, baromyringitis, rupture) and PET × URI interaction matrix; no a priori risk-tier cutoffs | Yes |

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| 12 | Methods — Development vs. validation | For validation, identify any differences from the development data in setting, eligibility criteria, outcome, and predictors. | §2.8 — Italian Air Force cohorts share altitude target and Teed-grade outcome; cohorts differ in (a) tympanometry pre-screening (tightened cohort prior applied) and (b) chamber profile detail (envelope reconstruction from published descriptions) | Yes |

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| 13a | Results — Partici-pants | Describe the flow of partici-pants through the study, including the number of partici-pants with and without the outcome and, if applicable, a summary of the follow-up time. | §2.8 (n = 7,271 exposures, n = 173 events, per-exposure rate 2.38%; Wilson 95% CI 2.06–2.75%); Italian AF cohorts: Morgagni 2010 n = 1,241; Morgagni 2012 n = 314; Landolfi 2009 n = 335 | Yes |
| 13b | Results — Partici-pants | Describe the charac-teristics of the partici-pants (basic demo-graphics, clinical features, available predictors), including the number of partici-pants with missing data for predictors and outcome. | §2.8 Predictors and Missing data (no individual-level demo-graphics required for cohort-aggregate fit) | Yes |

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| 13c | Results — Participants (validation only) | For validation, show a comparison with the development data of the distribution of important variables (demographics, predictors and outcome). | §3.2 Table 1; Italian AF cohorts vs FAC anchor — per-exposure rates and Wilson CI overlaps | Yes |
| 14a | Results — Model development | Specify the number of participants and outcome events in each analysis. | §3.1 Calibration (FAC anchor 173 / 7,271; achieved 2.47%); ABC-SMC 100 particles × 4 generations × 150 synthetic-cohort subjects per particle | Yes |

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| 14b | Results — Model development | If done, report the unadjusted association between each candidate predictor and outcome. | Mechanistic model — unadjusted associations are not estimated; predictor effects are encoded in §2.5 multipliers and §2.7 hazard model | Not applicable |
| 15a | Results — Model specification | Present the full prediction model to allow predictions for individuals (i.e., all regression coefficients, and model intercept or baseline survival at a given time point). | §2.3–§2.7 fully specify the model; full implementation released as open-source code at https://github.com/strikerdlm/barotrauma_model (tagged release v2.2.1-manuscript) | Yes |

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| 15b | Results — Model specification | Explain how to the use the prediction model. | §2.2 Model architecture; clinical-output description in §2.7 and §4.3; tutorial in repository documentation | Yes |
| 16 | Results — Model performance | Report performance measures (with CIs) for the prediction model. | §3.1 Calibration (FAC anchor 2.47% vs target 2.38%; ABC-SMC 95% CrI on rbarotitis); §3.2 External validation (Table 1, Figure 3, Wilson 95% CIs on each cohort); §3.4 Sobol indices with magnitudes | Yes |

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| 17 | Results — Model-updating (validation only) | If done, report the results from any model updating (i.e., model specification, model performance). | Not applicable — no model updating during external validation | Not applicable |
| 18 | Discussion — Limitations | Discuss any limitations of the study (such as nonrepresentative sample, few events per predictor, missing data). | §4.1 Limitations (calibration-anchor publication pending; conservative rupture threshold; modeler-judgment parameters; lumped-parameter muscle mechanics; published-envelope chamber-profile reconstruction) | Yes |

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| 19a | Discussion — Interpretation (validation only) | For validation, discuss the results with reference to performance in the development data, and any other validations. | §3.2 and §4 — external-validation results vs FAC calibration anchor; two of three Italian AF cohorts within Wilson 95% CI without refitting | Yes |
| 19b | Discussion — Interpretation | Give an overall interpretation of the results, considering objectives, limitations, results from similar studies, and other relevant evidence. | §4 Discussion paragraphs (aperture-collapse model load-bearing for chamber rates; PET four-state model novelty; URI temporal modifier as direct encoding of rhinovirus-challenge evidence) | Yes |

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| 20 | Discussion — Implications | Discuss the potential clinical use of the model and implications for future research. | §4.3 Conclusions (per-exposure quantitative risk for pre-chamber screening; recommended safe descent-rate ceilings for PET and URI patients) | Yes |

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| 21 | Other — Supplementary information | Provide information about the availability of supplementary resources, such as study protocol, web calculator, and data sets. | §2.1 + §Data availability and §Code availability statements on title page — open-source simulator at https://github.com/strikerdlm/barotrauma_model; aggregate cohort statistics in Tables 1–2 and SDCs 2–3; individual-level data on request from FAC institutional authority | Yes |

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| 22 | Other — Funding | Give the source of funding and the role of the funders for the present study. | Title-page Conflicts of Interest and Source of Funding — no external funding; institutional resources of DIMAE, Colombian Aerospace Force; funders had no role in design, analysis, writing, or submission decision | Yes |

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**Compliance summary:** 28 of 28 items compliant (with 2 items marked Not Applicable per TRIPOD's stated allowance for mechanistic-physics models without machine-learning components: 14b — no unadjusted predictor associations; 17 — no model updating during validation).

**Reference:** Collins GS, Reitsma JB, Altman DG, Moons KGM. Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD): the TRIPOD Statement. *Ann Intern Med.* 2015;162(1):55–63. doi:10.7326/M14-0697.